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Dreamcatching: introducing a reflexive tool to facilitate situated complexity in urban co-design practices

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ABSTRACT

The co-design of urban places and the role of professional urban designers entails facilitating multidimensional processes to capture people's dreams of attractive and inclusive future living environments. Nevertheless, many urban designers tend to focus on the material factors rather than the social and cultural ones. Based on findings from a case study and conceptual analysis, a systemic reflexive tool is offered to help scholars and practitioners to comprise the complexities of urban design. The Dreamcatcher tool highlights four perspectives of situated complexity and the synergies between them: place and setting; people and participation; vision and scope; and tools and methods.

KEYWORDS

Urban design; Dreamcatcher tool; situatedness; co-design; situated complexity; systemic design

Introduction

Urban design is broadly seen as a visionary process of creating better places than what exists today (Carmona et al. 2010). But better for whom and how? The underlying motivation for the current study is the observation that the cultural and social aspects of human everyday urban experiences are somewhat marginalized in urban design practices. One reason for this can be one of the dominant paradigms in design, that is, the techno-rationalist problem-solving approach, as outlined, for example, by Simon (1996). This approach aims to fix agendas, set goals, and design suitable courses of actions as rationally and effectively as possible, while aiming to change an existing situation into a preferred one. Yet, the techno-rationalist approach is critiqued for only being applicable in fixed situations with set goals and for neglecting to address the human dimension (Dorst 1997). Today, however, design challenges are increasingly complex and multidimensional (van der Bijl-Brouwer and Malcolm 2020). In the context of urban design, complexity can, for example, involve taking human cultural and social aspects into account to create better living environments (Hårsman Wahlström, Kourtit, and Nijkamp 2020) and not only deal with the functionality and materiality of the built environment. To do this, urban designers need to consider the nature of the systems in which both the situated place and they themselves are embedded (Drew, Robinson, and Winhall 2021). The latter ideas connect to another design paradigm: reflective practice (Schön 1983), where designing is seen as a process of interaction between the designers and the design situation by taking the whole system into account. Reflective practice involves reflecting

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on the consequences of actions, aiming to further knowledge and learning. It corresponds to Argyris and Schön's (1975) theories of single- and double-loop learning. Single-loop being a practice in which errors and mistakes are treated, but it does not fundamentally change anything else. Double-loop learning, on the other hand, involves reflectively questioning basic assumptions with the aim of transforming the practice. Designing in this paradigm can be seen as a process of handling ill-defined problems by performing situated actions. In such situations, it matters where you are when you do what you do (Clancey 1997). In line with Castello (2014) and Craig (2001), this paper defines situatedness as the designer's meaning-making activities connected to environmental/material, social, and cultural factors in relation to a specific place and project. In order to create places that 'foster positive human experiences and nurture our humanity' (Appleyard 2022, 39), there is need for further knowledge of the challenges connected to these issues, as well as knowledge of how to address them.

In design, there is knowledge of systemic, co-creative, multidimensional processes that aim to facilitate people's dreams of better futures (Sanders and Stappers 2008; Sleeswijk Visser 2009; Buchanan 2019). In this paper, the concept of dream catching originates from Sanders (2002), stating that to understand people's experiences, their relations with past and present, memories, and dreams is important. Thus, dream catching involves exploring people's values, motivation, aspiration, fears, memories, visions, wishes, feelings, and so on and so forth. Dream catching may sound ambiguous, and the techno-rationalistic design paradigm tends to avoid such subjective views. In Sweden, this has resulted in urban design activities organized in silos, separating human cultural and social aspects from city planning and design (Larsson and Jalakas 2013; Sim 2019). The lack of holistic views and transdisciplinary practices, however, risk resulting in the urban design practice being rationalized into decisions about technology, materials, and other physical/material entities, as Larsson and Jalakas (2013) point out. Such 'bounded rationality' (Simon 1996) is obviously one way of dealing with complexity. This paper, however, draws on Stolterman's (2008) argument that complexity can be seen as prosperous, including endless possibilities for new positive experiences, and that urban design complexity is a real and practical problem that needs to be dealt with by applying reflective design approaches.

Consequently, the aim of this paper is to explore what situated urban design complexity entails and how it may be utilized to serve the design processes. In the upcoming sections, systemic design and situatedness is discussed to outline a basis for an understanding of urban design complexity. A conceptual framework analysis and co-creative design research project are portrayed to further nuance the constitutes of situated complexity and to describe the iterative process that resulted in the main contribution of this paper, the Dreamcatcher tool. The tool aims to provide means to reflect upon, understand, and translate urban complexities into concrete action possibilities, to encourage urban design practitioners to cultivate situated design actions, and thus further contribute to knowledge of reflective urban design practices.

Situated systemic design

Systems knowledge has been central in design theory and practice since its scholarly beginning, however, with a renewed interest in later years (Buchanan 2019). Systems in

design should be understood 'as an organic whole of units or elements that form a whole and function together, working in unison to accomplish a human purpose' (Buchanan 2019, 98). In this view, there is a need to address the whole system whether it relates to the problem framing, the end-products, the design approach, the strategies, the situatedness, or the economic, social, and cultural interdependencies that should be addressed in theory and practice.

> Every product is a system of parts working together to accomplish a common purpose, whether in the graphic display of typography, images, colour, and pattern in a poster: the integrated workings of a physical artefact; the sequence of planned activities, communications, and exchanges of a service or any other human interaction; or the complexities of dynamic and evolving organizations, environments, and systems. (Buchanan 2019, 86)

These conceptions of systemic design are similar to how situatedness has been characterized by Simonsen et al. (2014):

> [t]o say that design is situated is to highlight the interactions and interdependencies between designers, designs, design methods, and the use situation with its actors, activities, structures, particulars, and broader context. (Simonsen et al. 2014, 1)

In these perspectives, design is described as relational, and the synergies and interconnectedness between the various factors are said to be significant and, in a way, inevitable. Another example of this is (Carvalho and Goodyear 2018) framework for design and learning networks consisting of the three perspectives: set design, which is *physically* situated to a place, a space, an artefact, or a tool; *social* design that is situated in relation to people, roles, or organizations; and *epistemic* design, which is situated by tasks and actions and how they are approached. Furthermore, the need to study designers and their environments as integrated systems (Craig 2001) highlights the complexity of urban design practice that involves taking numerous objectives, ideas, and viewpoints into account simultaneously. Thus, systemic design requires a set of skills that goes beyond the traditional design skills and mindset (Drew, Robinson, and Winhall 2021). Systemic designers, in this view, need integrative thinking, abductive reasoning, value judgement, reflexivity, and synthesis through prototyping and making. They also need to adapt to new roles that require visionary, social, and entrepreneurial skills.

Urban co-design as situated complexity

As discussed so far, systemic design can be defined as situated in various ways. There is no one fixed definition of 'situated'. Haraway (1988) coined 'situated knowledges' by stating that all knowledge is situated and partial, taking place in certain historical, political, and situational conditions. Therefore, it is proposed to ground knowledge claims by describing how they are locally and historically interrelated. In relation to design, Suchman (2002) proposes that plans are resources for actions, but should be seen as a set of 'situated actions' that can only be acted on in relation to the actors and the situation at hand. This is also in line with Simonsen et al. (2014), who state that all design is carried out from an embedded position, usually involves many participants, and encompasses a range of interactions and interdependencies among designers, designs, design methods, and users. Also, Lave and Wenger (1991) argue for 'situated learning', meaning that effective

learning can never happen if it is separated from the social practice in which it is supposed to be applied.

Urban design has its own subject-specific situated and interrelated dimensions, for instance: *morphological* (built form, structure, and patterns); *perceptual* (value and meaning); *social* (safety, power, equality); *visual* (aesthetics); *functional* (movement, activity, and affordances); and *temporal* (cycles of time and change) (Carmona et al. 2010). Furthermore, urban design takes place on various scales, from whole cities to specific neighbourhoods and places within them, and involves different stakeholders, which further adds to the situated complexity. Several approaches have been developed and applied previously to make sense of various aspects of urban design situated complexity. For example, Lynch's (1960) seminal work, *The Image of the City*, promotes the importance of organizing and designing cities in terms of paths, edges, nodes, districts, and landmarks to help people move and orient themselves. Another pioneering approach was introduced by Gehl (1987) in *Life between the buildings*. The latter focuses on how the built environment can create desirable conditions for necessary, recreational, and social activities to enhance people's quality of life. These days the widely applied multi-disciplinary practice of placemaking, which originates from the 1960s, builds upon the ideas of empowerment and the real-life collaboration between professionals, local residents, and decision-makers (Strydom, Puren, and Drewes 2018). In a similar manner, more recent concepts of place innovation and inclusive urban design practices address unequal power structures in terms of knowledge, economical resources, status, and connections related to, for example, ethnicity, gender, sexuality, and class, in urban design practices (Lindberg, Gelter, and Karlberg 2017; Wikberg Nilsson et al. 2022).

Method

Research approach

The methodological approach for this study is based on the conceptual framework analysis, proposed by Jabareen (2009). As the name indicates, conceptual framework analysis aims to construct new concepts and frameworks within a specific topic based on a variety of sources, such as articles, books interviews, and practices within the explored field. The initial literature search included academic journals, books, and electronic databases such as Scopus with keywords such as: situated design, situatedness, wicked problems, ill-defined problems, design practice, design thinking, designerly thinking, systemic design, systems design, urban design, and place design. The sources consist of books, academic refereed papers, and reports. The reason for this is that the initial research concerned how authors frame the scope of situatedness, systems, and complexity in design, and due to the publication norms of the field during some of the time, the sources were mainly books. Following the 1960–80s, more and more scholarly design journals discuss these issues. Also, different design organizations seem to have different means to share design knowledge. Some use reports or web sites. For the scope of this article, sources concerning computer systems design and engineering design have been left out. Later, the literature review was refined by detecting relevant sources from within the books, articles, and reports in the initial list. This is by no means meant to be a comprehensive review of all current literature on these issues, however, it

communicates some of the perspectives of how complexity, systemic perspectives, co-creation, and situatedness are addressed within design theories and practices. In addition, findings from a case study are included in order to deepen the analysis and understanding of how these perspectives can be addressed in the urban realm.

Case: people-tools-process-place

The three-year (2019–2022) design research project People-Tools-Process-Place set out to explore design for social inclusion in the urban realm. The idea was to implement a design-driven place innovation process in collaboration with two Swedish municipalities: Skellefteå and Mölndal. The aim was to contribute to knowledge of sustainable, inclusive, and attractive living environments by 1) co-creating situated urban design methods and tools, and by 2) co-creating a place in each of the participating municipalities. The cities of Skellefteå and Mölndal were chosen based on being two middle-sized Swedish cities, with a population of 73,200 and 69,350 respectively, one situated in the north, and the other in the south-west part of Sweden. In Sweden, large cities usually have resources in order to use consultants and/or various existing, rather costly, tools, checklists, and guides. Such tools are very different in their design and content and require different amounts of resources to be able to be successfully used. A challenge for small- and middle-sized cities is that they claim to not have competences or resources to perform such work. Sometimes the existing tools are simply not perceived as easy to use in a situated context, leading to arbitrary assessments and decisions. It can also be problematic to take care of the results in a good way, that is, transform the plans into practical actions in the development of a particular place.

Swedish politic for designed living environments stipulates that the urban design practice should consider how it impacts the way people live, work, and play, and to what extent people experience public places as inclusive and attractive (Swedish Government Offices 2018). Although most Swedish municipalities are aware of this, they have trouble in transforming political visions into concrete action possibilities in urban design projects. There is also a tendency to treat these ideas as isolated questions for other sectors in the municipality than the city planning offices to take care of. The current research project idea, hence, was to use the design process' pro- and intra-active explorations of existing and future living environments to reflect on the underlying principles and values that guide planning, analysis, and design of urban places. In addition, this should be done by probing methods and tools in the co-creation of two attractive and inclusive places, with participating actors from the municipalities, as a basis for double-loop learning.

In the current case, the research approach consisted of project meetings, workshops, and interviews, combined with literature reviews of areas identified as relevant for the project, for example, social and cultural aspects of the built living environment. Various professionals in the broader field of urban design, such as architects, landscape architects, planning engineers, traffic engineers, experts in accessibility, city communication office practitioners, social workers, community building practitioners, and also people knowledgeable in art in public spaces, participated in a series of workshops and interviews. In these, tools that aim to support the practitioners to reflect on place development were developed and tested. Workshops and interviews took place both in physical settings and



Figure 1. The participants collaboratively developed various design materials in various workshops that later resulted in the Dreamcatcher tool. Picture shows an early phase of the process (now-phase). Photo: Åsa Wikberg Nilsson.

online, due to pandemic restrictions. The on-site workshops were documented through field notes, and the digital workshops were recorded, with permission from all attendees, for use in the research project. The research material hence consists of field notes, audio and video tracks as well as the materials developed in the workshops, such as notes and drawings (see Figure 1). The workshop series were built around the concept of now-wow-how, inspired by the Future Workshop approach developed by Jungk (1987) and Jungk and Mullert (1989) to enable transition from fixed urban design practices and experiences into new ideas and action possibilities (Müllert 2017).

Outcomes

This section includes the findings from the conceptual framework analysis followed by an overall introduction of the main contribution of this paper, the Dreamcatcher tool and its reflexive questions (see Figure 2).

Situated perspectives of urban co-design

Urban co-design can be viewed from various situated, but interconnected perspectives: people and participation; vision and scope; place and setting; and tools and methods.

People and participation

Only when the urban environments are created by everybody, they can provide something for everybody, as urban activist Jane Jacobs famously once stated (Jacobs 1961). Participation can be considered – and occurs – on many different levels, ranging from pure manipulation to total citizen control (Arnstein 1969). The central question here is: are the places designed *for* or *with* people? In the case study, inclusion and participation were identified as crucial in achieving the long-term goal of attractive and sustainable societies.

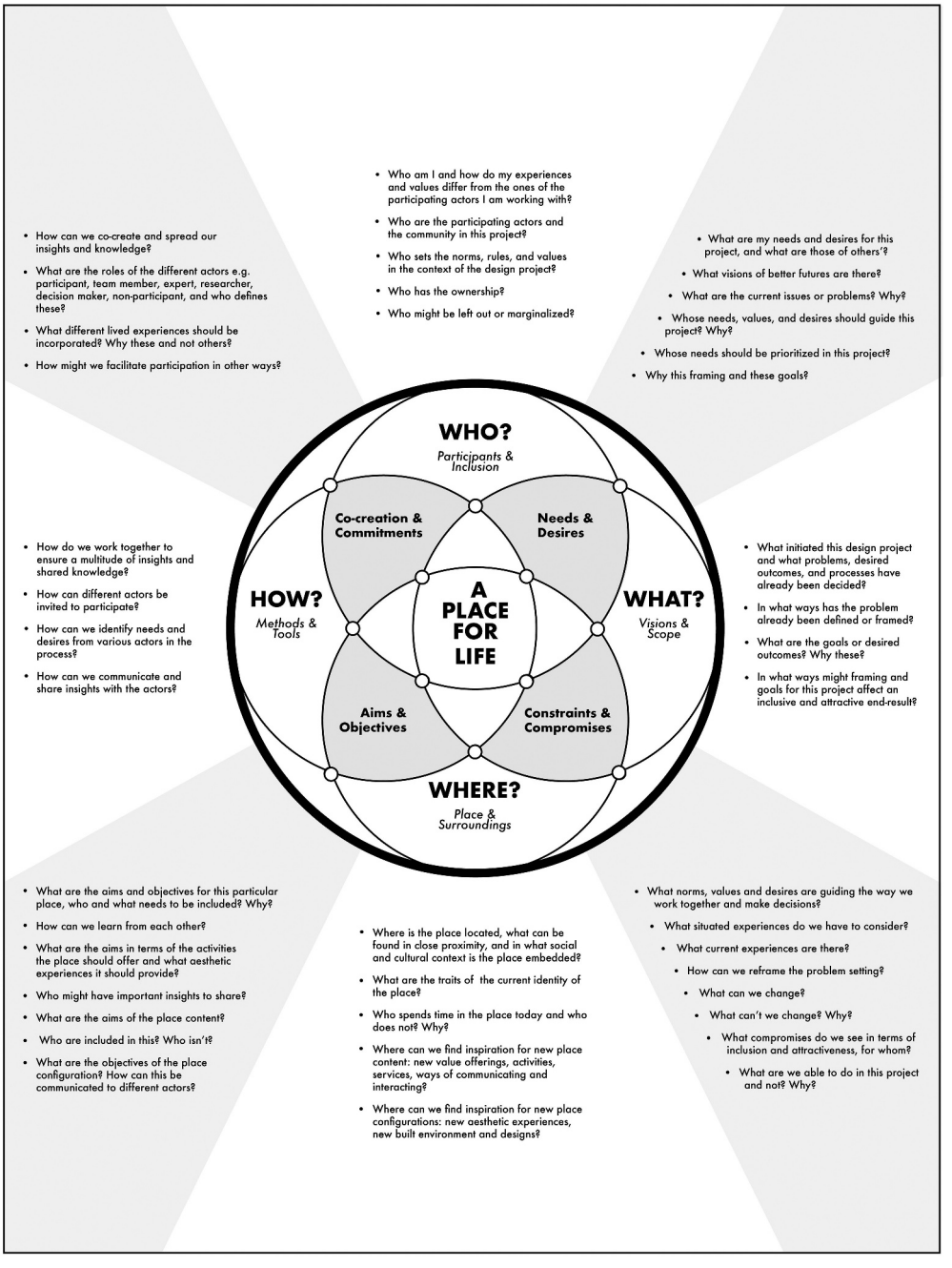


Figure 2. The Dreamcatcher tool aims to embrace and explore the rich situated complexity in urban co-design projects through set of questions. Illustration: Wikberg Nilsson & Eronen.

The challenge, however, lies partly in the concepts of inclusion and participation, as in understanding what both inclusion and its opposite, exclusion, actually mean, and for whom. Most often, it is not conscious discrimination that is behind the latter, but obliviousness of other perspectives or experiences than ones' own. But, as the

professional urban designers also might be everyday users of the places they design, they cannot be seen as objective or disconnected. They have a stake in the case. Furthermore, designers, such as other citizens, are a diverse group with many different backgrounds which influences both their actions and their preferences. The participants in the case study expressed that they see themselves as inclusive persons and aim at acting in non-excluding ways. Nevertheless, they discussed and exemplified occasions when unaware actions made them act in a non-inclusive manner without realizing it. For instance, one participant described a previous playground project where they (the place developers) thought they had acted with the intention of inclusion and attractiveness. However, when they visited the playground in question, people at the scene pointed out things they had not thought of.

As previously noted, it matters where you are when you do what you do, *and* it also matters where you come from and what you consciously or unconsciously choose to act on (Clancey 1997; Haraway 1988). For example, one identified fundamental principle is to feel safe in a place. In both Skellefteå and Mölndal, participants discussed safety as a necessity, but also challenging in terms of how it can be actuated for different people. The participants had their own notions of being safe at a particular place, but how could they develop empathy with the notions of others? What could the notion of being safe mean for different people: a child, an elderly person, a man, a woman, someone who is new at the place, for someone walking with a stick, or someone with social phobia, and what social and material aspects of the places could contribute to such experiences? The issue is complex and perhaps not all places need to, or can, have inclusive-for-all elements. Nevertheless, the importance of discussing the issues in relation to different peoples' experiences in each design project was recognized, as it is one part of, and contributes to, long-term attractive and sustainable places.

Vision and scope

Vision and scope are closely connected to frame creation that helps to explore the problem area and coordinate various ideas (Dorst 2015). Thus, one key feature in frame creation is to embrace complexity: 'If the scope of attention cannot be widened to a rich and complex field, no new frames can be created' (Dorst 2015, 103). New frames are important since they allow people to move on from the initial understanding of the task to more profound levels. In the context of urban design this means, for instance, that if the shaping of built environment is seen as the main task, it limits the visionary possibilities of what immaterial values the built environment contributes to. Thus, successful frame creation is a reflexive practice that involves various people and their repertoires – their [professional] skills, capabilities, values, understandings, previous experiences, and the feel for the situation (Schön 1983). This was also noted in the case study: throughout, the participants expressed the value of meeting both colleagues and other actors they normally do not collaborate with. They especially appreciated the possibility to collaborate early on in the design process and discuss various dimensions of inclusion and attractiveness from micro (individual) to macro (society/community) level.

Also, in the activities, the participants used cards with different user scenarios as a means to discuss different perspectives. These scenarios were combined with questions with the aim of initiating a reflection of what was needed for these users in the particular

place, with inspiration from Wikberg Nilsson and Jahnke (2018) and Wikberg Nilsson (2022). Some of these questions were later included in the Dreamcatcher tool.

Place and setting

Urban places include both material and immaterial dimensions (Carmona et al. 2010). Furthermore, places are not only situated in geographical locations, but they are also embedded in certain culture and history that change over time (Cresswell 2015). Larger places such as cities and neighbourhoods are challenging to rearrange, but other aspects are constantly developing and evolving. New buildings, roads, vegetation, and services appear constantly and both festive and everyday events take place over time. These changes also impact the experienced place identity and atmosphere (Adams et al. 2020). After returning from living elsewhere, several participants in the case study described how they noticed a positive difference in their cities in terms of, for example, values and attitudes. Such changes can be understood in terms of synergies between the built environment (configuration), the activities, services, and experiences (content), and the digital and on-site communication and marketing of the place that both form and are formed by the identity of a place in terms of its geography, history, values, and resources (Lindberg, Gelter, and Karlberg 2017). The resources and strengths of a place are thus seen as a significant basis for its further development. However, it is equally important to acknowledge the possible negative aspects in order to be able to reduce or transform them (Berleant 2011). Those issues were also addressed by the participants in the case study. They reflected on questions such as: how do we differ from our neighbouring cities, and how might the city centre also integrate and reflect the surrounding villages? These reflections surfaced as strengths, weaknesses, and challenges connected to the place, such as surrounding nature, existing infrastructure, and inhabitants' different needs, experiences, and expectations.

In the beginning of the project activities, there was a tendency among the participants to discuss the place as rather one-dimensional, often focused mainly on the synergies created by morphology and functionality. For example, the participants used top-down maps of the city when they discussed new place-making projects. For this reason, one of the project activities involved making eye-level images of the place, to enhance a varied understanding of the place itself. The reason for applying images and not actual site visits was the ongoing pandemic. The participants were also asked to bring an image of a place that they felt had the identity of the city itself, a strong 'sense' of the cities of Skellefteå and Mölndal, respectively. The activity was then to discuss what formed that experience, and if and how it could be identified in the materiality of the place.

Furthermore, the discussions often revolved around the legibility of the place (Lynch 1960). A central discussion was for this reason how to create design elements that could help people 'read the place', whether that was visually or with other senses, based on the different user scenarios presented. Based on the latter, the participants also discussed what the place should offer, in terms of different value offerings for different people. Gehl's (1987) approach was addressed, for instance, in terms of creating seating and transparent facades to enhance the recreational and social aspects, however, in the beginning there was a predominantly functional flare to the discussions. At the other end of the scale, it was discussed that the places should also contribute to sensual

experiences and well-being qualities that could be free from – or related to – pure functional aspects.

Methods and tools

In co-design, the involved actors are considered as experts on their own experience; they are skilled and knowledgeable in different ways, which also means that the actors can learn from each other (Sanders 2002; Sanders and Westerlund 2011; Sanders and Stappers 2008). Also, according to Manzini, all people have the capability of being a designer, by:

> / . . . /putting their design capacity into action: a way of thinking and doing things that entails reflection and strategic sense, that calls us to look at ourselves and our context and decide whether and how to act to improve the state of things. (Manzini 2015, 1)

This means that different individuals have their own unique points of view and understandings, and sharing those is important, as they frame the mutual understanding, process, and place simultaneously. As Haraway puts it: 'The only way to find a larger vision is to be somewhere in particular' (Haraway 1988, 590). However, in order to reach into the larger vision, the particular viewpoints need to be sewn together. This requires a shift of the professional urban designer activities from being an interpreter of people's needs into being a facilitator in a process of shared explorations (Sanders and Stappers 2008).

The participants in the case study found collaborative sessions valuable in terms of learning from each other and being nudged into new perspectives of their work. Some participants working in the field of urban design explicitly expressed that their regular workday involved using tools such as tables, diagrams, and numbers, while others applied visual methods such as sketching, photographs, and tangible models. This further illustrates the importance of increasing the awareness of utilizing different tools, especially in participatory activities with the citizens, in order to enhance inclusion and the degree of participation (Zamenopoulos et al. 2021; Hollander 2012).

The Dreamcatcher tool and its constituents

Design can be seen as a process of solving problems through reflective conversations with the situation and by using different materials to think with (Schön 1983). The reflective dialogues in this view involves reviewing the situation through an initial consideration, in which the situation starts to 'talk back'. This can be seen as a circular process, in which loose threads can be woven together. Likewise, it has been said that understanding happens when someone is interpreting something, through conversations with others in an iterative and hermeneutic cyclic process (Gadamer 1975).

This was the starting point when the aforementioned project participants initiated and tested several versions of a tool to start an urban design process, through asking reflexive questions of the situation and – not least – each other. Designers often use questions as a way of framing the problems in a design situation. For example, Simonsen et al. (2014) propose the four project-design questions of: analysis, what is the problem and for whom; construction, what technologies are available as building blocks; process, how can the design be accomplished; and vision, what is the solution? There is also the widespread, but not evidence-based, 5 Whys method, which assumes that the apparent cause of

a problem is often not the real issue, but rather something that needs further exploration in order to understand relations and consequences (c.f. Interaction Design Foundation n.d.; Design Thinking Methods Catalogue n.d.; Think n.d.). One of the project outcomes is hence the Dreamcatcher tool, consisting of four situated perspectives that are central in urban design: people and participants; vision and scope; place and setting; and tools and methods. These were developed into easy-to-grasp and fundamental questions to ask in urban design projects: 1) *who*: the involved participants' individual perspectives and competences, 2) *where*: the specific place and its surroundings, 3) *what*: visions and scope of the project in question, and 4) *how*: tools and methods for the particular project (see Figure 2). The need for clarifying sub-questions was highlighted in early user testing of the model where the participants expressed concerns that the tool risked remaining too conceptual. The sub-questions connected to each category were identified and formed during the project activities as well as inspired by the literature studies.

The use and validation

The overall objective of the Dreamcatcher tool is to help navigate discussions of different understandings and action-possibilities in relation to a place-specific project. The idea is that the facilitated conversations will help the participants comprehend the various complex interrelations. There is neither a specifically defined starting point for the conversations, nor is it necessary to deeply dive into all the questions posed in the tool. It is, however, essential to address all the areas to attain a holistic perspective. Based on the insights gained in the case study, it is proposed to begin the conversations by addressing the questions connected to the base constitutes of who, what, where, and how. This helps to create an understanding of the fundamental issues and provides a more stable ground for the more complex level, explored through the questions in the overlapping areas.

Therefore, in line with reflective practice (Schön 1983), the main idea is that by asking reflexive questions in relation to specific topics, assumptions will be made visible which in turn will help to make more informed decisions based on increased awareness, empathy, and commitment for the various issues related to a specific place and its users and other stakeholders. In other words, the dream catching activity involves surfacing hidden or taken for granted aspects, documenting and analysing them to create an understanding of what a better future might be, and how it can be realized.

The participants in the current project, in which the tool and its reflexive questions were developed during the three years, confirm that the tool can be used in different phases of a project:

> The tool works well as a start of for discussions, and when framing problems and seeking potential solutions. (Participant A, Architect, 3 March 2021)

The reflexive questions related to the tool were also experienced as helpful when evaluating design suggestions:

> We went through all of the points, one after the other, and this resulted in a change of some parts of the design plan./ . . . /We also see a need for continuous follow-ups, when you start to look at actual heights, surrounding buildings and other aspects you have to change them in accordance to the specific site./ . . . /We also discussed where people want to sit to take a break, to have different outpost where people can overlook the place./ . . . /To create

> a 'room-in-the-room' in the large space in front of the main entrance, we added a pergola with lights. (Participant B, Urban design project manager 10 March 2021)

The tool has also been tested in other contexts than the current case, such as a project aiming to develop an attractive region. Although those participants initially experienced the tool somewhat complex, they validated that it is beneficial to have questions that cover various areas as point of departure, since it allows an initial broad exploration instead of immediately focusing on possible solutions. They also reported that questions posed in the tool generate new questions, which enables the users to modify the tool according to their specific situation.

The Dreamcatcher tool is intended to be applied in conjunction with other methods and tools, as a supplement to existing activities. The user tests and feedback show that the tool can be used in the beginning of a project as a base for discussing and understanding the specific place from various perspectives, during the project to deepen the understanding and evaluate design ideas with different participants and stakeholders, and after the project is finished to reflect on and harvest learnings for upcoming projects. The themes and the reflexive questions were confirmed as important to address, and it was seen as valuable to have a tool that combined social/cultural aspects with material/physical properties. However, the ultimate validation of the Dreamcatcher tool will be to see that it is still in use in urban design practices in upcoming years and to see in the long run that it contributes to better living environments for all.

Lastly, some considerations regarding the name 'dreamcatcher' since it may be perceived as controversial. It emerged during analysing the research material to understand how the different aspects might relate to each other. The visuals resembled a traditional dreamcatcher that in some Native-American and First Nations cultures is seen as a magical object that can capture dreams and block out nightmares (Harris 2019). As beforementioned, in the field of design, the core of the design activity is often claimed to be peoples' wishes, dreams, and needs, also called desiderata – an inclusive whole of something that is wanted, ought to be, and needs to be (Nelson and Stolterman 2012). Similarly, marketing research shows the need to shift focus from exploring people's use experiences to their dream experiences in order to reach innovative outcomes in terms of societal challenges (Dube, Helkkula, and Strandvik 2014). Thus, the choice of the word 'dreamcatcher' does not seek to undermine or misappropriate any other understanding and use. On the contrary, it was considered a suitable positive metaphor for exploring different people's aspirations for a better future in relation to a particular place or living environment. Also, the project participants in the case study validated it as relevant and meaningful. Naming the tool Dreamcatcher aims to emphasize the need to reduce people's bad design experiences and address their hopes and desires also in the field of urban design.

Discussion

The incentive for this paper was an observed need for urban designers to respond to situated complexity. As identified in the case study and in the literature review, most urban designers possess great knowledge, but the knowledge is often fragmented into different specializations which makes it challenging to employ in a systemic way. Also, it was noted that the human dimension of urban design is seen as demanding to address

knowledgeably and responsibly in urban design practices. In the case study, material factors connected to the built environment were experienced as complex, but easier to deal with than the social and cultural factors that influence the experience of a place. Addressing both human and material dimensions at the same time was challenging for the current project participants.

As noted in the beginning of this paper, the overall challenge is to address the techno-rationalistic problem-solving practice that at least in Sweden seems to dominate the public sector and many consultancies. The lack of means to address the situated complexity risks the process of urban design being rationalized into decisions of technology, materials, and other physical entities and not addressing human experiences in and of the place (Larsson and Jalakas 2013). To avoid this risk, the reflexive Dreamcatcher tool is proposed. The reflexive questions involve finding strategies to question one's own attitudes, thought processes, values, and habitual actions, that is, to understand our complex roles in relation to others. The Dreamcatcher tool hence offers potential to address several levels: a macro level that relates to what kind of design paradigms to apply in order to create better future communities and a sustainable society as a whole, a meso level that relates to participation and ownership of a place, and a micro level that connects to different individuals and their various needs of a place.

Firstly, on a *macro-level*, the Dreamcatcher tool addresses questions that traditionally have been applied in urban design practices. In addition, it also integrates the reflective practice paradigm (Schön 1983) through considering the particulars of the people, the place, and the process. In the model this is visualized through reflexive questions of the *who*, the participants and their understanding and different standpoints; the *where*, as in discussing the place and its specific surroundings; the *what*, the visions and scope as the frame to reflect upon; and the *how*, as in thinking about the methods and tools that should be applied. Such knowledge can be used by urban designers to ensure actions that address both environmental/physical, social, and cultural factors.

Secondly, on a *meso-level*, it addresses the need for collaborative activities in the urban design practice. Who should participate, and how could they participate in creating the better futures? Complex problems will never be completely solved, but the rich possibilities they include can be collectively dreamed and explored (Drew, Robinson, and Winhall 2021). A visual tool may facilitate participants' discussions of the various synergies and support in discovering the areas previously not addressed, and thus guide the designers in their search for understanding the specific needs of various people connected to a particular place (Bresciani 2019). Furthermore, the tool encourages including individuals with other values or lived experiences than held by those who are in power positions, that is, people working in municipalities planning/design offices, architecture practices, and so on.

Thirdly, on a *micro-level*, the Dreamcatcher tool challenges participants' own beliefs and actions especially through the reflexive questions. Situated complexity involves the interwoven functions of a designers' meaning-creating activities in relation to their own positionality (e.g., Castello 2014; Craig 2001; Haraway 1988). In the case study it was observed that the activities and the reflexive questions showed potential in making the participants consider and articulate the rationale behind design choices. Sometimes, these reasons were mainly connected to the functional aspect of the built environment, such as removal of snow or maintenance. This can be explained as a single-loop reflection

(Argyris and Schön 1975) that does not involve any change. Other times, the activities and reflexive questions made the participants re-think their own actions and choices. It is proposed that this happens when the activities, design material, and questions confront the participants with things they have not done or thought of before, and hence force them into a double-loop reflection cycle. Reasons that guide actions and set the priorities can be considered as values. Making the values that guide the design processes visible opens a path for the possibility to challenge and nudge them into new ways of thinking and doing. In a metaphorical sense this can be seen as reducing 'the nightmares': avoiding design outcomes that are based on values that correspond neither to anyone's needs nor their desires. Also, increasing awareness of the consequences of the design choices and the possible compromises they may include enables the urban design practitioners to better communicate with citizens, which was a need expressed by the participants in the case study.

As the aim of this paper was to explore what situated urban design complexity entails and how it may be utilized to serve the design processes, further research in various settings is needed in order to explore how well the Dreamcatcher tool serves its intended purpose to advance co-design of attractive living environments. Future studies need to focus on the values, methods, and means to guide a reflexive situated urban design process through all the complexity it entails.

The concepts and their synergies visualized in the Dreamcatcher tool help to navigate situated complexity in urban design practices and provide nuanced insights and inspiration to guide the considerations of multiple aspects and diverse futures. In summary, the tool enables:

- *Holistic perspectives*: where people's experiences, everyday aesthetics, and collaboration are important factors in the creation of a future attractive and inclusive built environment.
- *Inclusive and democratic perspectives*: to achieve an inclusive and democratic society with access to good living environments, based on both particular and common needs.
- *Knowledge development*: situated learning among participating actors.

Consequently, it is proposed that attractiveness in relation to urban living environments increases by reflecting, embracing, and exploring the rich situated complexity as a potential source for catching dreams, rather than problems.

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